

Biochemical Markers in Colorectal Cancer: Diagnostic and Therapeutic Implications

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Colorectal cancer is the second most common cancer in men and women in the United States. In 1988, colorectal cancer will account for approximately 147,000 new cancer diagnoses and 62,000 deaths.⁴² Despite active and ongoing research into its treatment, the survival of patients with colorectal cancer after surgical resection has not changed over the past 40 years.⁴⁵ It appears appropriate to devote more attention to potential methods of early cancer detection and/or prevention.

On the assumption that early diagnosis may improve survival, research has been directed at identifying markers in the serum or colonic mucosa of patients with colorectal cancer, preferably in early stages, but not present in patients without colorectal cancer.⁴¹ Early diagnosis is the key issue, since this is a stage at which the cancer is potentially curable.⁸ The current emphasis is on developing biochemical and genetic markers for the diagnosis of patients at risk for developing colorectal cancer as well as for the early diagnosis of potentially curable disease.

CARCINOEMBRYONIC ANTIGEN

Carcinoembryonic antigen (CEA), initially described in 1965,¹² is the first and most widely used plasma tumor marker. CEA belongs to a class

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of "oncofetal proteins," which are normally found in fetal tissue but only in miniscule or undetectable amounts in normal adult tissue. CEA is a group of related high molecular weight glycoproteins whose immunoreactivity resides on the protein moieties. It has been and is still currently the standard against which other oncofetal antigen tumor markers are measured. Unfortunately, its sensitivity and specificity for diagnosing early colon cancer are not high. Nevertheless, it may have a role in the postoperative monitoring of patients after resection of colorectal cancer to determine if surgical extirpation of cancerous tissue is complete and to follow the patient for recurrent disease.^{10, 13}

Although initially thought to be specific for gastrointestinal malignancies, it is also elevated in other nondigestive tract cancers and in many nonmalignant conditions including liver disease, renal failure, fibrocystic breast disease, and in smokers.^{10, 12, 52} In addition, CEA has very low sensitivity for the detection of early resectable colorectal cancers and no sensitivity at all for adenomatous polyps, which are the ultimate premalignant colorectal lesion.¹⁰ Therefore, CEA is not useful as a screening test in asymptomatic patients.

Adding to the clinical problems with CEA is the lack of a good functional definition of the range of CEA in normal subjects. If one defines a CEA elevation as only a substantial increase (such as values greater than 10 ng per mL), then the specificity of CEA is greatly increased, but its sensitivity is concomitantly sacrificed.

Currently, there are primarily two commercially available methods for measuring CEA. The CEA-Roche is the older assay and requires acid extraction of plasma before CEA determination by radioimmunoassay. The CEA-Abbott assay is an enzyme-linked immunoassay (ELISA). Although the two methods produce parallel results in the majority of cases, there is inter-assay variability. The two tests should not be used interchangeably in monitoring an individual patient, and the particular assay method should always be identified in the test report.¹¹

With these caveats, large population studies have shown that patients with colorectal cancer have significantly higher CEA levels than patients without colorectal cancer. Unfortunately, the overlap is very large, and the diagnostic value of CEA for the individual patient is lost.⁵²

CEA should therefore not be used in the screening of asymptomatic individuals. In one study of over 2000 adults, 73 were found to have elevated CEA values. Of these, only nine were found to have cancers, but seven of those were already clinically symptomatic at the time. In the greater than 2000 subjects with normal CEA values, 25 patients were found to have digestive tract cancers. This confirms the futility of using CEA as a screening test.⁷

Evidence suggests that preoperative levels of CEA may correlate with subsequent recurrence, and the time to recurrence, of colorectal cancers.⁴⁸ Unfortunately, advanced stage colorectal cancer is often poorly differentiated and hence produces low amounts of CEA, and the prognostic value of CEA is lost.¹¹

Where elevated CEA levels are produced by the colorectal cancer, the resection of such cancers may result in a decrease in plasma CEA levels

over a 4- to 8-week period. The assumption is then made that recurrences are more likely to exhibit similar CEA-producing capacity as the initial cancer. Thus, CEA has been used to monitor the completeness of cancer resection, response to cancer therapy, and recurrence.⁴⁸ A new elevation or an ongoing rise in plasma CEA is assumed to be due to tumor recurrence, and second-look surgery has been advocated for these patients.⁴⁸ The reported experience with this proposed strategy has been controversial.

In a series from the Mayo Clinic, 166 patients with Dukes's stage B2 or C colorectal cancer were followed after a potentially curative resection. CEA was found to be elevated to greater than 5.0 ng per mL in 33 of 48 patients with recurrent cancer, with an apparent sensitivity of 67 per cent. Yet, 37 of the 118 patients without recurrent cancer also demonstrated a CEA level of greater than 5.0 ng per mL, showing the specificity to be less than 50 per cent. If a CEA level of greater than 10.0 ng per mL was chosen as a cutoff, then the specificity of the test increases to 100 per cent, but the sensitivity declines even further. An additional problem was that the recurrences detected by CEA monitoring tended to be distant metastases, for which there is no effective treatment.⁴ In this series, 28 of 36 patients (75 per cent) with distant metastases demonstrated a high CEA, whereas only 6 of 13 patients with local recurrence (46 per cent) showed a high CEA, but even the locally recurrent cancers were *not* resectable in *any* of the six cases.¹⁵

In an analysis of three published series, 100 patients on CEA surveillance, 5 to 15 per cent, eventually came to "second-look" surgery, but only 2 to 8 per cent had resectable tumor recurrence and only 1 to 3 per cent were disease free after a follow-up of a "few years."¹⁰ In another study, plasma CEA was measured every 2 to 3 months in 583 patients with potentially curative resection of digestive tract cancer (more than 75 per cent had colorectal cancer). A rise in CEA occurred 7 to 10 months before the cancer was detectable on clinical grounds. The recurrent tumor was resectable in 46 per cent of patients whose recurrence was detectable solely by a rising CEA.⁴³ Among all patients who came to surgery for recurrent cancer in this study, surgery was inexplicably delayed 4 to 8 months after the initial CEA elevation.⁴³ Earlier surgery might have improved resectability and survival rates.

Proponents of CEA monitoring emphasize that in order to be effective, plasma CEA must be measured as frequently as every 6 to 8 weeks. In the Mayo Clinic Series, CEA was measured "at least every 15 weeks," and Staab et al⁴³ measured CEA every 2 to 3 months. In two series in which CEA was monitored every 6 to 8 weeks or more frequently, some success has been reported.^{31, 33}

A multicenter trial reported that among 130 patients with recurrent cancer, 105 (80 per cent) had a CEA greater than 2.5 ng per mL, whereas only 6 patients of 2760 without recurrence had a CEA greater than 2.5 ng per mL. Seventy-five patients underwent surgery. In 43 patients, a high CEA was the sole indication of recurrence, and the recurrent cancer was resectable in 23 of these patients (59 per cent). However, recurrent cancer was resectable in only 50 per cent of those patients whose recurrence was detectable clinically.³³ Fifteen of the 39 patients who had complete resection

remained disease free, but the proportion of patients who went to surgery specifically because of a high CEA and who remained disease free after resection was unknown. Patients with recurrent disease who had plasma CEA monitored every 1 to 2 months had a higher 5-year disease-free survival (13 of 39; 33 per cent) than those patients with recurrence in whom CEA was monitored less often.³³

This same group have also published their own experience at their medical center (Ohio State University), and reported similarly good results.³¹ Beginning in January 1976, plasma CEA was monitored every 4 to 6 weeks using a nomogram to validate CEA elevations, and 124 asymptomatic patients underwent second-look surgery solely because of an elevated or rising CEA. Among patients with recurrent cancer discovered at the second-look operation, and in whom 5-year follow-up was available, the 5-year survival after second operations was 14 of 45; 31 per cent).³¹

Such a relatively successful experience with CEA needs to be confirmed by prospective trials that include a control group not subjected to CEA monitoring. At the present time, CEA monitoring should not be considered standard practice. If it is carried out, the CEA level should be checked every 6 to 8 weeks for the first 8 to 12 months following a potentially curative resection. Furthermore, such CEA monitoring should only be performed in an organized fashion within the framework of a well-designed and executed clinical trial. Finally, even the proponents of CEA agree that it is of relatively little value in those patients who had normal CEA values before resection of their colorectal cancer.

Other investigators have attempted to use CEA in combination with other tumor markers with little success. CEA was evaluated together with ten other biochemical markers in malignant and benign colonic epithelial disease. CEA in combination with four of the most sensitive markers increased the sensitivity in detecting early disease; however, the specificity was decreased by 30 per cent.³² More investigation into the combined use of multiple biochemical markers needs to be done.

Radiolabelled monoclonal antibodies to CEA have been injected into patients and subsequent scintiscanning has been performed to look for occult colorectal cancer. However, the sensitivity and specificity of this technique is less than that of CT.²

Currently, CEA is still the best serum marker available for diagnosing colon cancer. However, because of its low sensitivity for early disease, its usefulness is limited. Additional clinical trials are needed to evaluate its sensitivity and specificity for detecting treatable recurrent disease following curative surgery.

ABNORMAL GLYCOPROTEIN

Many cell surface lipids and proteins undergo posttranslational modifications that consist of glycosylation and phosphorylation. Many of these modifications are known to alter protein function and are important in cell transport, cell receptors, and intercellular interactions.^{9, 16} Changes in the glycosylation patterns of premalignant and malignant colonic cells have

been identified. Many of the aberrant carbohydrate structures seen on tumor cells are also seen in fetal tissue. These abnormal carbohydrate moieties have implications not only for the pathogenesis of colorectal cancer, but also for the detection of early disease.

CA-19-9 is a monoclonal antibody to a glycolipid in tissues, but recognizes a mucin-type glycoprotein in serum.³⁰ It was developed by using a human colon cancer cell line, SW1116, as an immunogen, and recognizes the carbohydrate moiety (a monosialoganglioside).^{22, 29} This antibody, although originally developed against colon cancer to aid in its diagnosis, is more sensitive for gastric, pancreatic, and hepatobiliary tumors as a serum marker.^{14, 18, 43} However, its usefulness is limited by its lack of specificity. Many benign tissues such as pancreas, gall bladder, and gastric epithelial cells show immunoreactivity.^{3, 18} Improved serological diagnosis of colon cancer may be achieved by the combination of antibodies that recognize different abnormal carbohydrate determinants. The combination of CSLEA1, an antibody that recognizes a monosialoganglioside similar to CA-19-9, with CSLEX1, an antibody to a sialylated Lewis^x antigen, was more sensitive than either was alone in the serologic diagnosis of colorectal cancer.⁶ For serologic detection of colon cancer, CEA was found to be more sensitive than CA-19-9, although CA-19-9 was more sensitive in pancreatic cancer.¹⁴

Many of the currently studied carbohydrate cancer antigens are similar to carbohydrates in human blood group antigens. CA-19-9 is a sialylated Lewis^a antigen. As mentioned earlier, CSLEX1 is a sialylated Lewis^x antigen. The Lewis antigen, like other blood group antigens, is found only in erythrocytes in a small percentage of persons designated as nonsecretors. Some of these nonsecretors have been found to express blood group antigens in their malignant colorectal tissue.⁴⁰ Many of these blood group antigens are abnormally expressed in colorectal cancer also in secretors, who normally express their antigens in tissues other than erythrocytes.

The normal expression of blood group antigens in the colonic mucosa is limited to the proximal half of the colon in patients with a blood group secretor status.^{40, 50} The only exception is the Lewis^a, which is expressed throughout the entire colon and is independent of secretor status.⁴⁰ In fetal colonic epithelial tissue, all antigens are expressed throughout the colon.^{40, 50} In colorectal cancer and some adenomatous polyps, the expression of blood group antigens in the neoplastic tissue and surrounding normal mucosa is altered in several ways. These alterations in blood group antigen expression include (1) fetal blood group antigens are expressed in distal colon lesions; (2) blood group antigen expression is lost in normal colon tissues; (3) blood group antigens are expressed in nonsecretors; and (4) blood group antigens incompatible with the host are also expressed. These alterations in blood group antigen expression may be of potential use in early cancer detection, but their significance or use is unclear at present.

The advent of monoclonal antibody technology has allowed us to explore further abnormal blood group antigen expression in colorectal cancer. Colorectal neoplasms express not only normal blood group antigens in an altered manner as described earlier, but they also express altered blood group antigens.^{1, 18-20} These altered blood group antigens include

polymeric Le^x and their sialylated derivatives, and Le^y, a fucosylated derivative of Le^x.^{1, 18-20}

These antigens are also considered "oncofetal antigens" because they are expressed in fetal colonic epithelium but not normally expressed in adult tissue. Their expression in adenomatous polyps closely correlates with malignant potential, that is, the larger the polyp and the more villous component present, the more likely these antigens are expressed.^{1, 18} The T antigen (Thomsen-Friedenreich antigen), a precursor carbohydrate for the blood group system M and N antigens, has also been studied in colorectal tissue by immunohistochemical localization.⁵ Currently, these are only research tools but may eventually be developed as potential diagnostic markers in high-risk populations.

ORNITHINE DECARBOXYLASE

Ornithine decarboxylase (ODC) is a key enzyme in the synthesis of polyamines from arginine and ornithine. The precise function of the polyamines is unknown; however, much experimental data suggest that they are very important in many cellular functions, including cell proliferation, differentiation, and neoplastic transformation.^{36, 37, 47} ODC and the polyamines have been shown to be important in cell proliferation in intestinal mucosa: (1) during recovery after cytotoxic injury,²⁸ (2) during lactation,⁴⁹ (3) during postintestinal resection adaptation,²⁶ and (4) during newborn maturation³⁷ in a rat model. Both ODC and polyamines have been demonstrated to be elevated in many malignant tissues including colorectal cancer.^{27, 34} Data from our laboratory and others suggest that ODC and polyamines are elevated in tissues that progress to malignancy and they may serve as a marker of tissues predisposed to malignant transformation.²⁵ Finally, experimental evidence in animal models suggests that decreasing the level of ODC and polyamines may be effective in cancer chemotherapy³⁵ and potentially also in chemoprevention in high-risk groups.

Studies in our laboratory suggest that measuring ODC in the colonic mucosa of persons at risk for developing adenomatous polyposis coli may serve as a marker to identify those who will later express manifestations of multiple colonic polyposis and cancer.²⁵ ODC activity was measured in the colonic mucosa of patients with adenomatous polyposis coli, stratified according to gross and histologic criteria from normal-appearing flat mucosa to dysplastic polyps, and compared with normal controls. There was a statistically significant increase in ODC activity from flat mucosa to polyp to dysplastic polyp. We have now found that ODC activity in colorectal cancers is another threefold higher than seen in the dysplastic polyps.

Furthermore, in colonic mucosa from clinically unaffected first-degree relatives of patients with adenomatous polyposis coli who have a 50 per cent chance of inheriting the gene, there was a bimodal distribution of ODC activity. One peak occurred at the mean for normal controls and the other was near the mean for the normal-appearing mucosa from patients with adenomatous polyposis coli.²⁵ In the 4 years since that study, three persons with high ODC have developed multiple polyps. None of the 17

persons with low ODC have developed polyps. Although further follow-up is necessary, the data suggest that measuring ODC activity in colonic mucosa may identify patients who have inherited adenomatous polyposis coli before the development of multiple polyps.

Other investigators have also shown elevations in ODC in adenomatous polyps and cancers from patients with sporadic colorectal cancer. In one study, a significant increase in the ODC activity of polyps and colorectal cancers compared with normal control mucosa was found.³⁹ In another study, the ODC activity in 34 colorectal carcinomas and six polyps were elevated compared with uninvolved mucosa in the same patients. The ODC activity in the normal mucosa of patients with cancer was also elevated compared with the mucosa from normal subjects without cancer or polyps.²³ Additional studies have documented a trend in increasing ODC activity from normal-appearing mucosa to polyp and ultimately to colon cancer.³⁸ While ODC is not specific for colorectal cancer, it may serve as a biochemical marker for mucosa at risk for malignant change.

THERAPEUTIC IMPLICATIONS

ODC and the polyamines may serve not only as potential diagnostic markers for colorectal cancer, but, because they are important for cell proliferation, they may also provide potential targets for the treatment and/or prevention of colorectal neoplasia. Currently, inhibitors of polyamine synthesis are available.³⁵ Difluoromethyl ornithine (DFMO), a specific inhibitor of ODC, has been shown to be well tolerated with few side effects, and has been used to suppress tumor growth in vitro and in vivo.³⁵ DFMO has been used in many laboratories, including ours, to suppress mucosal growth processes in rats,^{26, 28, 49} suppress carcinogen-induced colonic tumors in mice and rats^{21, 39} as well as the growth of implants of human colon carcinoma cells in athymic mice.¹⁷ DFMO has also been documented to inhibit tumor metastases.⁴⁶

DFMO has limited efficacy in the treatment of established cancers due to an incomplete inhibition of the polyamine pathway.³⁵ Other polyamine synthesis inhibitors and antagonists are currently being tested for use in cancer chemotherapy. Because of its low toxicity and side effects, DFMO may still be useful in the chemoprevention of patients at high risk for developing colorectal cancer. DFMO may be useful in adenomatous polyposis coli patients who have undergone colectomy with ileal-rectal anastomosis because of their continued risk of developing cancer in the retained rectum. Preliminary studies in these patients have shown that DFMO can effectively suppress colonic mucosal ODC activity and all proliferation as measured by thymidine incorporation.²⁴

SUMMARY

Markers that are now in use, including CEA and CA-19-9, are not specific or sensitive enough to detect early colorectal cancer. Newer tumor

markers such as polyamines, ornithine decarboxylase, and altered blood group carbohydrate antigens may have a potential as future tumor markers. Additional studies of these markers as well as the development of new biochemical markers are warranted in the future to enhance the sensitivity and specificity of diagnosis of early colorectal cancer and those at risk for developing cancer. Finally, understanding events involved in abnormal cell proliferation (that is, elevated polyamines and ODC in colorectal cancer) may help direct future chemotherapy and possibly chemoprevention in high-risk groups such as adenomatous polyposis coli.

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